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*
*          STAAD.Pro
*          Version 2007    Build 04
*          Proprietary Program of
*          Research Engineers, Intl.
*          Date=    APR 15, 2013
*          Time=    11:26:29
*
*          USER ID: Parsons Brinckerhoff
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1. STAAD SPACE

INPUT FILE: vent duct VER2.STD

2. START JOB INFORMATION

3. ENGINEER DATE 27-NOV-12

4. END JOB INFORMATION

5. INPUT WIDTH 72

6. UNIT METER KN

7. JOINT COORDINATES

8. 1 0 0 0; 2 3.75 0 0; 3 0 5 0; 4 0 5 3.5; 5 3.75 5 0; 6 3.75 5 3.5

9. 7 0 0 3.5; 8 3.75 0 3.5; 9 0 8 0; 10 0 8 3.5; 11 3.75 8 0

10. 12 3.75 8 3.5; 13 0 11 0; 14 0 11 3.5; 15 3.75 11 0; 16 3.75 11 3.5

11. 17 0 14 0; 18 0 14 3.5; 19 3.75 14 0; 20 3.75 14 3.5; 29 0 12.5 0

12. 30 3.75 12.5 0; 31 0 12.5 3.5; 32 3.75 12.5 3.5; 33 3.75 2 0

13. 34 3.75 2 3.5; 35 0 2 0; 36 0 2 3.5; 37 0.5 5 0; 38 0.5 5 3.5

14. 39 0.5 8 0; 40 0.5 8 3.5; 41 0.5 11 0; 42 0.5 11 3.5; 43 0.5 14 0

15. 44 0.5 14 3.5; 45 0.5 2 0; 46 0.5 2 3.5; 47 3.25 5 0; 48 3.25 5 3.5

16. 49 3.25 8 0; 50 3.25 8 3.5; 51 3.25 11 0; 52 3.25 11 3.5; 53 3.25 14 0

17. 54 3.25 14 3.5; 55 3.25 2 0; 56 3.25 2 3.5

18. MEMBER INCIDENCES

19. 1 1 35; 2 2 33; 3 3 37; 4 3 4; 6 7 36; 7 8 34; 8 4 38; 17 3 9; 18 5 11

20. 19 9 39; 20 9 10; 21 11 12; 22 4 10; 23 6 12; 24 10 40; 25 10 3; 26 9 4

21. 29 12 5; 30 11 6; 33 9 13; 34 11 15; 35 13 41; 36 13 14; 37 15 16

22. 38 10 14; 39 12 16; 40 14 42; 41 14 9; 42 13 10; 45 16 11; 46 15 12

23. 49 13 29; 50 15 30; 51 17 43; 52 17 18; 53 19 20; 54 14 31; 55 16 32

24. 56 18 44; 57 18 13; 58 17 14; 61 20 15; 62 19 16; 95 5 6; 104 53 41

25. 105 43 51; 116 29 17; 117 30 19; 118 31 18; 119 32 20; 120 33 5

26. 121 6 33; 122 34 6; 123 5 34; 124 35 3; 126 36 4; 129 35 45; 131 36 46

27. 132 33 34; 133 37 47; 134 38 48; 135 39 49; 136 40 50; 137 41 51

28. 138 42 52; 139 43 53; 140 44 54; 141 45 55; 142 46 56; 143 47 5

29. 144 48 6; 145 49 11; 146 50 12; 147 51 15; 148 52 16; 149 53 19

30. 150 54 20; 151 55 33; 152 56 34; 153 54 42; 154 44 52; 155 51 39

31. 156 41 49; 157 52 40; 158 42 50; 159 49 37; 160 39 47; 161 50 38

32. 162 40 48; 163 47 45; 164 37 55; 165 48 46; 166 38 56

33. START GROUP DEFINITION

34. JOINT

35. _COLBASE1 1 2 7 8

36. MEMBER

37. _COL1 1 2 6 7 17 18 22 23 33 34 38 39 49 50 54 55 116 TO 120 122 124 126

38. _LBEAM1 20 21 36 37 52 53

39. _LBEAM1A 4 95 132

40. _LOADBEAM 3 8 19 24 35 40 51 56 129 131 133 TO 152

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41. _ANGLE1 104 105 153 TO 166
42. _ANGLE2 25 26 29 30 41 42 45 46 57 58 61 62 121 123
43. _SPlice 49 50 54 55
44. _LOADBEAM1A 3 8 19 24 35 40 51 56 129 131
45. _LOADBEAM1B 143 TO 152
46. END GROUP DEFINITION
47. MEMBER PROPERTY JAPANESE
48. _COL1 TABLE ST H414X405X18
49. _LBEAM1 TABLE ST H150X150X7
50. _LBEAM1A TABLE ST H150X150X7
51. _LOADBEAM TABLE ST H150X150X7
52. _ANGLE1 TABLE ST C250X90X11
53. _ANGLE2 TABLE ST C250X90X11
54. MEMBER RELEASE
55. _ANGLE1 START MX MY MZ
56. _ANGLE1 END MX MY MZ
57. _ANGLE2 START MX MY MZ
58. _ANGLE2 END MX MY MZ
59. _LBEAM1 START MY MZ
60. _LBEAM1 END MY MZ
61. _LBEAM1A START MY MZ
62. _LBEAM1A END MY MZ
63. _LOADBEAM1A START MZ
64. _LOADBEAM1B END MZ
65. DEFINE MATERIAL START
66. ISOTROPIC MATERIAL1
67. E 2.05E+008
68. POISSON 0.3
69. DENSITY 77
70. ISOTROPIC STEEL
71. E 2.05E+008
72. POISSON 0.3
73. DENSITY 76.8195
74. ALPHA 1.2E-005
75. DAMP 0.03
76. END DEFINE MATERIAL
77. CONSTANTS
78. *BETA 45 MEMB _ANGLE1
79. MATERIAL STEEL ALL
80. SUPPORTS
81. 1 2 7 8 PINNED
82. LOAD 1 DL
83. SELFWEIGHT Y -1
84. LOAD 2 LL
85. MEMBER LOAD
86. _LOADBEAM1A CON GY -2.508 0.35
87. _LOADBEAM1B CON GY -2.508 0.15
88. LOAD 3 WX
89. MEMBER LOAD
90. _COL1 UNI GX 0.283
91. _LBEAM1 UNI GX 0.141
92. _LBEAM1A UNI GX 0.141
93. * WIND LOAD = 0.613 X 34 X 34 = 708.628 N/M2
94. * ACTING ON ONE JOINT OF THE VENTILATION DUCT = 3M X 1.5M X 708.628 /1000 = 3.1
95. _LOADBEAM1A CON GX 3.189 0.35
96. _LOADBEAM1B CON GX 3.189 0.15

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STAAD SPACE

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97. LOAD 4 WZ
 98. MEMBER LOAD
 99. _COL1 UNI GZ 0.283
 100. _LOADBEAM UNI GZ 0.141
 101. _LOADBEAM1A CON GZ 3.189 0.35
 102. _LOADBEAM1B CON GZ 3.189 0.15
 103. LOAD COMB 5 DL LL
 104. 1 1.0 2 1.0
 105. LOAD COMB 6 DL WX
 106. 1 1.0 3 1.0
 107. LOAD COMB 7 DL WZ
 108. 1 1.0 4 1.0
 109. LOAD COMB 8 DL WCOMB
 110. 1 1.0 3 1.0 4 1.0
 111. LOAD COMB 9 DL LL WX
 112. 1 1.0 2 1.0 3 1.0
 113. LOAD COMB 10 DL LL WZ
 114. 1 1.0 2 1.0 4 1.0
 115. LOAD COMB 11 DL LL WCOMB
 116. 1 1.0 2 1.0 3 1.0 4 1.0
 117. LOAD COMB 12 UDL LL
 118. 1 1.4 2 1.6
 119. LOAD COMB 13 UDL WX
 120. 1 1.0 3 1.4
 121. LOAD COMB 14 UDL WZ
 122. 1 1.0 4 1.4
 123. LOAD COMB 15 UDL WCOMB
 124. 1 1.0 3 1.4 4 1.4
 125. LOAD COMB 16 UDL LL WX
 126. 1 1.2 2 1.2 3 1.2
 127. LOAD COMB 17 UDL LL WZ
 128. 1 1.2 2 1.2 4 1.2
 129. LOAD COMB 18 UDL LL WCOMB
 130. 1 1.2 2 1.2 3 1.2 4 1.2
 131. PERFORM ANALYSIS

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 48/ 93/ 4

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 34/ 12/ 78 DOF
 TOTAL PRIMARY LOAD CASES = 4, TOTAL DEGREES OF FREEDOM = 276
 SIZE OF STIFFNESS MATRIX = 22 DOUBLE KILO-WORDS
 REQ'D/AVAIL. DISK SPACE = 12.4/ 258349.1 MB

132. LOAD LIST 1 TO 4
 133. PRINT SUPPORT REACTIONS